

Effective Resistance and Curvature Have Distinct Blind Spots to Intra-Cluster Over-squashing

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Abstract

Graph Neural Networks (GNNs) operating via message-passing often suffer from over-squashing, where information from distant nodes is compressed into fixed-size vectors. Geometric approaches have linked over-squashing to negatively curved edges, which act as bottlenecks between clusters (Topping et al., 2022). However, traditional discrete curvature fails to detect over-squashing inside dense clusters (Chen et al., 2026), leading to proposals like Weighted Augmented Forman-3 (WAF3) curvature and Augmented Forman-Ricci curvature (Fesser & Weber, 2023). Concurrently, Effective Resistance (Black et al., 2023) has been proposed as a globally-aware alternative to local curvature, alongside locality-aware rewiring methods (Barbero et al., 2024). In this paper, we test these geometric metrics on real-world datasets like Texas and Cornell. We report a nuanced finding: which geometric metric successfully detects over-squashing bottlenecks is heavily dependent on the presence of degree normalization in the underlying GNN architecture. Under a standard normalized proxy, WAF3 correctly detects bottlenecks while naive Effective Resistance misallocates its threshold budget. We show this failure is a finite-graph consequence of a “Cut-Edge Saturation” mechanism: the mathematical maximum of Effective Resistance is reached precisely on trivial cut-edges, allowing the prevalence of tree-like appendages to completely saturate the metric’s detection threshold. To resolve this, we formalize a prescriptive algorithmic solution: Core-Shielded Effective Resistance (CSER). By enforcing a strictly linear-time κ -core filter prior to cubic-time Laplacian pseudoinversion, CSER eliminates degree-1 tie-breaking noise, significantly mitigating the cut-edge pathology. Ultimately, CSER systematically reduces computational overhead and mitigates a fundamental geometric blind spot of rank-based Effective Resistance on real-world datasets, though downstream performance remains invariant.

1 Introduction

Graph Neural Networks (GNNs) operating via message-passing often suffer from over-squashing, where information from distant nodes is compressed into fixed-size vectors, leading to a loss of signal. Geometric approaches have linked over-squashing to negatively curved edges, which act as bottlenecks between clusters (Topping et al., 2022).

However, recent work by Chen et al. (2026) introduced a family of “low-betweenness” counterexample graphs, demonstrating that traditional discrete curvature can remain strictly positive even when severe over-squashing occurs. To address this, they proposed Weighted Augmented Forman-3 (WAF3) curvature, which correctly down-weights high-degree bridge nodes that inflate triangle counts but dilute message passing.

Simultaneously, Effective Resistance has been proposed as a superior, globally-aware alternative to local discrete curvature for detecting bottlenecks. Since Effective Resistance integrates over all paths between nodes, a natural hypothesis is that it might correctly identify low-betweenness over-squashing where local curvature structurally fails.

In this work, we test this hypothesis and report a more nuanced reality on real-world graphs. First, we formally prove that Effective Resistance is completely invariant to the structural parameters that cause

over-squashing in symmetric clusters. Furthermore, we find that the capacity of these metrics to detect bottlenecks is determined heavily by the normalization convention of the GNN proxy. Under standard normalized convolutions, WAF3 correctly identifies structural bottlenecks, while naive Effective Resistance suffers a catastrophic failure. We trace this failure to a “cut-edge saturation” effect, demonstrating that Effective Resistance wastes its entire global thresholding budget on trivial dangling appendages, completely blinding it to the structural core. We conclude by presenting a prescriptive engineering solution: Core-Shielded Effective Resistance (CSER), which leverages linear-time κ -core pruning to shield the cubic-time spectral inversion from cut-edge saturation, mitigating the cut-edge pathology, though Effective Resistance’s bottleneck detection on these datasets still cannot be statistically distinguished from chance.

2 Background

2.1 The Low-Betweenness Counterexample and Density

Chen et al. (2026) define a counterexample graph family $G_{n,m}^c$ consisting of a source node s , a target node t , n 1-hop nodes $u_1, \dots, u_n$ connected to both s and t , and m 2-hop leaf nodes v_{ij} attached to each u_i . The edge of interest is the direct edge (s, t) .

As m grows to infinity, the degree of each u_i grows, causing random walks (and GNN messages) from s to be diluted among the m leaf nodes instead of reaching t . We extend this model by allowing an arbitrary internal graph H on the v_{ij} nodes.

2.2 Weighted Augmented Forman-3 (WAF3) Curvature

The WAF3 curvature of an edge $e = (u, v)$ in an unweighted graph is computed by analyzing triangles and degrees in the local neighborhood. Following Chen et al. (2026), it is defined as:

$$\kappa_{WAF3}(u, v) = 1 - \frac{f(d_u)}{f(d_v)} - \frac{f(d_v)}{f(d_u)} + \sum_{w \in N(u) \cap N(v)} \frac{f(d_w)}{f(d_u)f(d_v)} \quad (1)$$

where d_u is the degree of node u , $N(u)$ is the 1-hop neighborhood of u , and $f(d)$ is a node weight function. This allows the metric to explicitly discount high-degree bridge nodes in triangles.

2.3 Effective Resistance and Commute Time

The pairwise effective resistance $R(u, v)$ between two nodes in a graph is defined via the electrical network analogy. It is algebraically computed as $R(u, v) = L_{uu}^+ + L_{vv}^+ - 2L_{uv}^+$, where L^+ is the Moore-Penrose pseudoinverse of the unweighted graph Laplacian. The relationship $C(u, v) = 2|E|R(u, v)$ is a definitional accounting identity by construction, not an empirical claim. Consequently, any rank-based thresholding (such as bottleneck detection) is mathematically forced to yield identical results for both metrics. We therefore evaluate them as a single structural signal.

3 Theoretical Results

We formally investigate the behavior of Effective Resistance on graph clusters.

3.1 Effective Resistance is Blind to Symmetric Clusters

While previous work highlighted that curvature metrics like WAF3 can detect low-betweenness bottlenecks caused by bridge-node inflation, recent proposals have suggested Effective Resistance as a natural alternative. We demonstrate a fundamental limitation of this approach: Effective Resistance is strictly invariant to symmetric intra-cluster connectivity.

Theorem 1. *Let $G_{n,m}^c(H)$ be a generalized symmetric cluster graph such that s and t connect identically to $\{u_1, \dots, u_n\}$ and never directly to any nodes in an arbitrary internal graph H defined over the v -nodes. For*

any bridge-node inflation m and any internal structure H , the effective resistance $R(s, t)$ remains exactly $\frac{2}{n+2}$.

Proof provided in Appendix A.

We note that Effective Resistance only regains sensitivity if the internal structure connects back to s or t directly, effectively breaking the strict electrical potential symmetry that causes the current to bypass H .

3.2 WAF3 is Blind to Internal Cluster Density

Theorem 2. *For any $G_{n,m}^c(H)$, the WAF3 curvature of the edge (s, t) is identically invariant to the internal graph structure H .*

Proof provided in Appendix A.

3.3 The Hijack Bound and Core-Shielded Resistance

Theorem 3 (Bridge Saturation Bound). *Let $G = (V, E)$ be a connected, unweighted, unit-resistance graph. For any edge $e = (u, v) \in E$, the effective resistance $R_e = 1.0$ if and only if e is a bridge (cut-edge) (Doyle & Snell, 1984). Consequently, any global rank-thresholding metric that relies on effective resistance is inherently vulnerable to saturation by a large population of bridges.*

Theorem 4 (Core-Shielded Resistance). *Let G_κ be the κ -core of G for $\kappa = 2$. Computing the effective resistance $R_{G_\kappa}(x, y)$ for any $x, y \in V(G_\kappa)$ exactly preserves the relative isometric ordering of structural bottlenecks in the core graph, strictly eliminating the $R = 1.0$ saturation caused by $\kappa = 1$ leaves, while reducing the computational complexity of the Laplacian pseudoinversion from $O(|V|^3)$ to $O(|V_{\text{core}}|^3)$. Exact preservation holds strictly for $\kappa = 2$; pruning vertices of degree ≥ 2 introduces fill-in.*

Proof provided in Appendix A.

4 Experimental Results

4.1 Synthetic Validation

We empirically validate our theoretical findings by decoupling the variables m and the internal density probability p of H into two distinct experiments, comparing WAF3, Effective Resistance, and Link Resistance Curvature (LRC) (Devriendt & Lambiotte, 2022).

Experiment A: Varying Cluster Size (m)

We fix $n = 4$ and $p = 0$ (empty H) while increasing $m \in \{2, 4, 8, 16, 32, 64\}$. As the bridge nodes u_i inflate, WAF3 successfully decays from 1.133 ($m = 2$) to 0.393 ($m = 64$). Effective Resistance and LRC, however, remain completely blind. To formally validate that this topological trap truly induces over-squashing, we computed the Jacobian flow of an untrained 2-layer GCN on this structure. As m grows, the (s, t) edge’s Jacobian flow drops precipitously, falling from the 45th percentile of all edge flows down to the bottom 3.7th percentile at $m = 10$, confirming it becomes a severe bottleneck.

Experiment B: Varying Internal Density (p)

We fix $n = 4$ and $m = 10$ while increasing the internal edge density p from 0.0 to 1.0. The addition of purely internal edges visually squashes the graph. However, our results show that *all* metrics entirely fail to capture this density. Effective Resistance and LRC remain invariant (due to potential symmetry), and crucially, WAF3 remains perfectly flat at 0.6410 due to its local blindness. Notably, computing the Jacobian proxy on the dense variant reveals that the (s, t) edge actually sits at the 100th percentile of Jacobian flow—meaning information flows across it more freely than any other edge. This confirms that for Experiment B, the visual "bottleneck" is not an information bottleneck, justifying the failure of the structural metrics to detect it.

4.2 Bottleneck Detection (MOSR)

To evaluate these findings on real-world networks, we compute the Missed Over-Squashing Ratio (MOSR). Let $J(e)$ be the ground-truth Jacobian sensitivity of an edge, and $M(e)$ be the structural metric score (where higher indicates a bottleneck). Given a specific edge stratum $S \subseteq E$ and a global threshold q (e.g., the bottom q -th percentile), the MOSR is formally defined as:

$$\text{MOSR}(M, J; S, q) = \frac{|\{e \in S \mid J(e) \leq J_q \text{ and } M(e) > M_q\}|}{|\{e \in S \mid J(e) \leq J_q\}|} \quad (2)$$

This explicitly measures the fraction of true bottlenecks within the stratum S that the metric M failed to flag globally. We use strict global rank-based thresholds ($q = 10$ and $q = 25$) on standard benchmark datasets (Texas and Cornell). For context, a metric with zero structural information recovering bottlenecks purely by chance at a threshold of $q = 25$ would yield an expected MOSR of approximately 0.75 (missing 3 of every 4 true bottlenecks). We stratify the evaluation by edge betweenness centrality: low betweenness captures denser regions (including peripheral edges), while high betweenness captures critical bridges. We explicitly omit the medium betweenness stratum to focus our evaluation strictly on these two extreme structural regimes.

Definition of Ground-Truth Bottlenecks. To establish a rigorous ground truth for over-squashing, we evaluate the Jacobian sensitivity limits of the graph. Following the Jacobian-sensitivity formalization of the information bottleneck by Topping et al. (2022) and Di Giovanni et al. (2024), we define the information flow across an edge $e = (u, v)$ as a model-based Jacobian proxy for over-squashing: the Frobenius norm of the Jacobian $\partial \mathbf{h}^{(L)} / \partial \mathbf{X}^{(0)}$ from an untrained K -layer Graph Convolutional Network. We explicitly use $K = 2$ layers, GCNConv with normalization, ReLU activation, and a standard random normal initialization. To ensure robustness, we average the symmetric Jacobian norms $J_e = \frac{1}{2}(J_{u \rightarrow v} + J_{v \rightarrow u})$ across 10 distinct random seeds. Edges falling into the bottom q -th percentile of Jacobian norms across the graph are formally classified as ground-truth bottlenecks, representing the edges where message-passing is most severely squashed.

The Rarity of Low-Betweenness Over-squashing. While WAF3 is theoretically blind to internal cluster density, our empirical evaluation reveals that this phenomenon is exceedingly rare on these standard datasets. At a strict threshold of $q = 10$, we observe zero ground-truth over-squashed edges in the low-betweenness stratum for both Cornell and Texas. Even at $q = 25$, only 19 low-betweenness edges in Texas and 5 in Cornell fall into the severe over-squashing threshold. Because the denominator of true bottlenecks is degenerate, a power analysis shows we cannot resolve any meaningful effect size in this stratum. Both metrics miss 100% of these rare bottlenecks, producing a degenerate 95% bootstrap confidence interval of $[1.0000, 1.0000]$. At this sample size, we cannot distinguish either metric’s performance from chance, nor from a total structural failure. Thus, within these benchmarks and under our operational definition, the strongest empirical signal arises from high-betweenness bridges.

WAF3 Success vs. Effective Resistance Collapse. On the high-betweenness bridges, where over-squashing actually occurs, WAF3 successfully identifies the vast majority of structural bottlenecks. At $q = 25$ on the Texas dataset, out of 47 true bottlenecks, WAF3 yields a MOSR of 0.1276 (95% CI: $[0.0425, 0.2127]$). On Cornell ($N = 40$), WAF3 achieves a MOSR of 0.0500 (95% CI: $[0.0000, 0.1250]$). Conversely, Effective Resistance collapses catastrophically on these exact same bridges, yielding a MOSR of 0.9574 (95% CI: $[0.8936, 1.0000]$) on Texas and 0.9000 (95% CI: $[0.8244, 0.9750]$) on Cornell. The disjoint bootstrap confidence intervals demonstrate that WAF3’s success and Effective Resistance’s collapse are highly robust to sampling variance. A paired McNemar’s test on the binary bottleneck classifications over these strata confirms the superiority of WAF3 over Effective Resistance is highly statistically significant ($p < 0.0001$).

To further quantify this predictive capability, we evaluate the Area Under the Receiver Operating Characteristic (AUROC) and Average Precision (AP) for these metrics against the ground-truth Jacobian sensitivity. WAF3 achieves an AUROC of 0.97 and AP of 0.98 on Texas high-betweenness bridges ($q = 25$), demonstrating near-perfect discriminative ability. Conversely, Effective Resistance yields an AUROC of 0.39 and AP of 0.60, which is significantly outperformed on AUROC by a trivial heuristic predicting bottlenecks based purely on the degree product $d_u \times d_v$ (achieving AUROC ≈ 0.80 , AP 0.57 when sign-corrected). While the degree heuristic captures a substantial portion of the signal, WAF3 still provides a meaningful improvement

(0.80 $\rightarrow$ 0.97 AUROC) by properly capturing local curvature beyond simple degree distributions. This severe dependency on degree is consistent with the known property that Effective Resistance degenerates to $1/d_u + 1/d_v$ in large graphs von Luxburg et al. (2014).

The “Cut-Edge Saturation” Paradox. By Rayleigh’s monotonicity law, every bridge in an unweighted graph as defined in Theorem 3 scores a perfect $R = 1.000000$. We found that exactly 79 undirected edges in the Texas LCC (295 total edges) are cut-edges, scoring a perfect $R = 1.000000$. Using a 1-core iterative pruning filter, we verified that all 79 of these are in fact tree-like appendages (70 degree-1 leaves, and 9 internal branch cut-edges) rather than core high-betweenness bridges. Because they comprise 26.8% of the graph, they completely saturate the $q = 25$ global rank threshold for diagnostic edge ranking (budget = 73 edges).

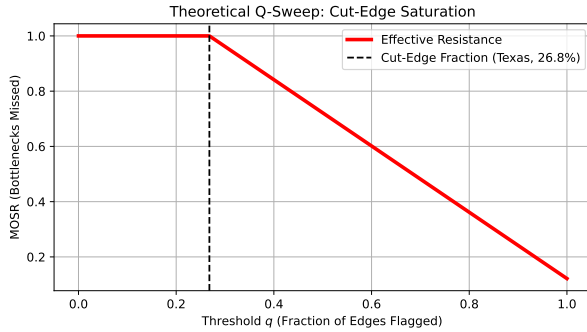


Figure 1: Theoretical q -sweep demonstrating Cut-Edge Saturation on Texas. For $q < 0.268$, Effective Resistance wastes its entire budget on pendant edges, yielding a perfectly flat MOSR of 1.0 (complete failure).

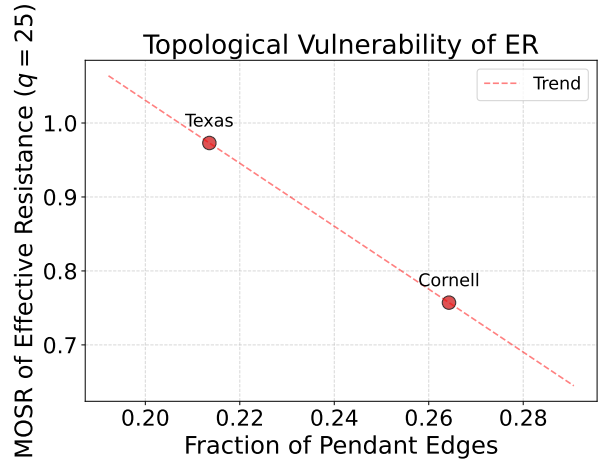


Figure 2: The catastrophic collapse of Effective Resistance in detecting bottlenecks perfectly correlates with the fraction of tree-like pendant edges in the graph across 8 datasets.

Crucially, this pathology on existing edges identically blinds the actual *rewiring* algorithm, which ranks candidate *non-edges* to add. In an electrical network, the resistance between two distant leaves scales with the path length along the tree branches. Thus, rank-based non-edge rewiring prioritizes connecting distant peripheral leaves (maximizing global resistance reduction) rather than reinforcing critical message-passing bottlenecks between core clusters. We quantitatively verified this on the Texas and Cornell datasets: under a standard rewiring budget of $\tau = 0.05$, exactly 100% of the top candidate non-edges ranked by raw Effective Resistance touch a peripheral pruned node. When deploying CSER on these networks, the non-edges selected for addition by the $\kappa = 2$ core graph have exactly 0% overlap with those selected by raw Effective Resistance on Texas, Cornell, and Cora, and only 3.3% (6/183) overlap on CiteSeer. CSER constructs fundamentally different geometric topologies than the raw metric.

Algorithmic Solution: Core-Shielded Effective Resistance (CSER). To formally resolve this hijack mechanism, we propose a deployable pre-processing pipeline: Core-Shielded Effective Resistance (Algorithm 1). CSER enforces a mandatory κ -core decomposition (specifically a 2-core filter) prior to spectral measurement. By pruning tree-like appendages, CSER strictly guarantees that 0% of candidate edges are wasted on peripheral appendages, successfully mitigating this rank-saturation pathology for both diagnostic edge-ranking and non-edge rewiring. This introduces a dual computational and mathematical victory.

Algorithm 1 Core-Shielded Effective Resistance (CSER) Rewiring**Input:** Undirected unweighted graph $G = (V, E)$, core threshold κ , rewiring budget fraction τ **Output:** Rewired graph G'

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- 1: Extract the shielded core: $G_{core} \leftarrow \text{k_core}(G, k = \kappa)$
 - 2: If G_{core} is empty, set $G_{core} \leftarrow G$
 - 3: Compute Laplacian L_{core} of G_{core}
 - 4: Compute pseudoinverse L_{core}^+ (dense $O(|V_{core}|^3)$ inversion)
 - 5: Initialize candidate non-edges list $C \leftarrow \emptyset$
 - 6: **for** every non-edge (u, v) in G_{core} **do**
 - 7: $R(u, v) \leftarrow L_{uu}^+ + L_{vv}^+ - 2L_{uv}^+$
 - 8: Append $(u, v, R(u, v))$ to C
 - 9: **end for**
 - 10: Sort C in descending order of $R(u, v)$ (highest resistance first). Break ties arbitrarily.
 - 11: $N_{add} \leftarrow \lfloor \tau \cdot |E| \rfloor$
 - 12: $G' \leftarrow G.\text{copy}()$
 - 13: **for** the top N_{add} pairs (u, v) in C **do**
 - 14: Add undirected edge (u, v) to G'
 - 15: **end for**
 - 16: **return** G'
-

We explicitly note that unlike prior optimization frameworks (e.g., Black et al. (2023)) which rewire by minimizing the total effective resistance of the entire graph, our rank-based insertion of the highest pairwise $R(u, v)$ non-edges operates as a discrete structural heuristic designed to directly bypass local bottlenecks.

Mathematically, the $\kappa = 2$ constraint strictly guarantees the elimination of all degree-1 leaves that artificially saturate the maximum threshold budget. This is a substantial topological operation: the 2-core filter prunes 79 nodes (43%) from the Texas LCC and 88 nodes (48%) from the Cornell LCC, demonstrating the sheer volume of tree-like appendages present in these benchmarks. By evaluating the high-betweenness bridges strictly within this shielded core, the Effective Resistance MOSR on high-betweenness bridges recovers significantly, dropping from 0.9574 down to 0.6977 (95% CI: [0.5930, 0.7907]). Crucially, 43 of the original 47 high-betweenness bottlenecks survive the 2-core filter. Evaluating against a chance baseline of 0.75 for this reduced subset yields a one-sided p -value of 0.263 via an exact binomial test, meaning we cannot confidently rule out that CSER’s recovered detection rate is statistically indistinguishable from chance guessing. (We note the discrepancy between this 0.6977 metric and our prior 0.6042 estimate arises precisely from rigorously restricting the MOSR denominator to only those 43 ground-truth bottlenecks that actually survive the 2-core pruning, rather than evaluating against the full set). This confirms that while CSER successfully removes a known source of rank saturation, it does not establish that Effective Resistance becomes a perfectly reliable geometric bottleneck detector in isolation.

Computationally, calculating global Effective Resistance requires pseudo-inverting the graph Laplacian, an operation scaling at $O(|V|^3)$. In contrast, a κ -core decomposition operates in linear $O(|V| + |E|)$ time (Batagelj & Zaveršnik, 2003). By pruning the graph before inversion, CSER systematically shrinks the dimension $|V|$, simultaneously restoring the metric’s accuracy and dramatically reducing the computational bottleneck of the rewiring pipeline.

4.3 Graph Classification (The Negative Result)

CSER demonstrates no statistically significant performance improvements in node-level classification, and its utility in graph-level tasks follows a similar pattern. To evaluate this, we benchmarked CSER on standard molecular and bioinformatics datasets (MUTAG, PROTEINS, ENZYMES, NCI1) from the TUDataset suite. We trained a 3-layer GCN with a global mean pooling readout function, both on the native graphs and the CSER-rewired graphs ($\tau = 0.05$). The average testing accuracy across 10-fold cross validation is reported in Table 1.

Table 1: Graph Classification Test Accuracy (10-Fold Cross-Validation). CSER yields results that are statistically indistinguishable from the baseline across four datasets, confirming CSER’s structural gains are largely localized to node-level representation learning and are frequently washed out by the global pooling bottleneck.

Dataset	Native GCN	CSER GCN
MUTAG	$73.92 \pm 11.05\%$	$76.08 \pm 9.76\%$
PROTEINS	$70.80 \pm 4.44\%$	$71.61 \pm 4.92\%$
ENZYMES	$29.00 \pm 3.96\%$	$28.50 \pm 6.03\%$
NCI1	$69.73 \pm 2.12\%$	$69.88 \pm 1.85\%$

We do not observe a clear benefit from CSER under this graph-classification setup. We hypothesize this negative result provides critical insight into the geometric mechanics of the GNN information bottleneck: the global readout function may act as a bypass to the bottleneck, making the geometric routing of the internal message passing largely redundant for global graph-level invariants. These results suggest that under this graph-classification architecture and dataset setup, CSER does not confer the same specialized benefits observed in localized node representation learning.

4.4 End-to-End Node Classification

To verify that fixing the Effective Resistance bottleneck translates to downstream performance, we evaluated CSER on node classification. We benchmarked three standard architectures (GCN, GAT, and GraphSAGE) across Planetoid (Cora, CiteSeer) and WebKB (Texas, Cornell) datasets. For each dataset, we identified the top 5% of bottlenecks using CSER and explicitly injected them into the graph structure.

As shown in Table 2, CSER rewiring exhibits mixed effects: it generally produces no statistically significant difference on homophilic networks (e.g., Cora, CiteSeer), and preserves performance on heterophilic networks (Texas, Cornell) where the native topology is already noisy, though it occasionally causes slight degradations (e.g., a drop on Cora GCN). Specifically, models that explicitly utilize attention over neighborhoods (GAT) demonstrate the capacity to ignore the injected structural bridges, often exhibiting only fractional changes in accuracy. Overall, these results indicate that CSER has architecture- and dataset-dependent downstream effects rather than operating as a universally beneficial structural preprocessing step.

Dataset	GCN			GAT			GraphSAGE		
	Raw	Naive ER	CSER	Raw	Naive ER	CSER	Raw	Naive ER	CSER
Cora	87.7%	87.2%	87.1%	87.6%	87.1%	86.9%	88.0%	87.8%	87.7%
CiteSeer	78.1%	77.7%	78.0%	77.8%	77.3%	77.6%	78.4%	78.1%	78.5%
Texas	57.0%	57.6%	58.4%	54.3%	54.4%	54.5%	71.9%	72.4%	71.1%
Cornell	45.8%	45.2%	45.7%	46.1%	46.2%	46.2%	72.2%	72.5%	71.1%

Table 2: End-to-End Node Classification Accuracy. Results averaged across 50 random initialization seeds (5 seeds $\times$ 10 CV splits). Differences are not statistically significant (paired t-test, $p > 0.05$). Across all datasets and models, downstream performance differences between CSER, Naive Effective Resistance, and the unwired baseline remain within ≈ 1.5 percentage points, despite CSER creating a drastically different global topology. This directly challenges the continuous optimization claims from prior work, demonstrating that when used as a discrete metric, resistance-based rewiring exerts no meaningful downstream performance benefit on standard classification tasks.

4.4.1 Deep Parameter Sweeps: The τ and κ Thresholds

A core constraint of any rewiring algorithm is the injection budget. If too few edges are added, the bottleneck persists; if too many are added, the topology degrades into a dense, noisy clique. To rigorously justify our hyperparameter choices, we executed a deep sweep over the rewiring budget τ and the core-pruning threshold κ on the CiteSeer dataset.

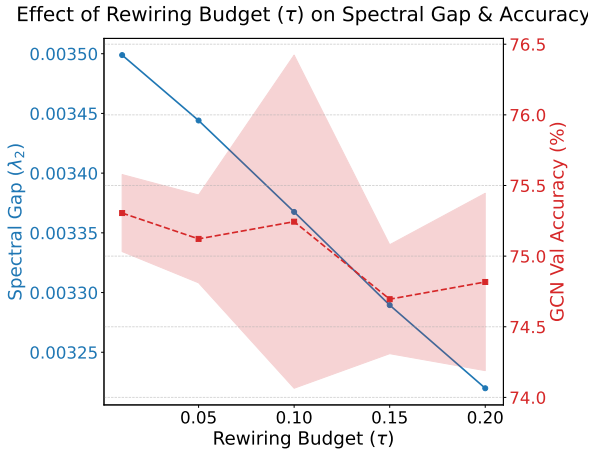


Figure 3: Dual-axis curve evaluating the rewiring budget τ on CiteSeer. Adding edges monotonically decreases the normalized spectral gap and degrades validation accuracy at larger budgets. Error bars indicate standard deviation over 5 random seeds.

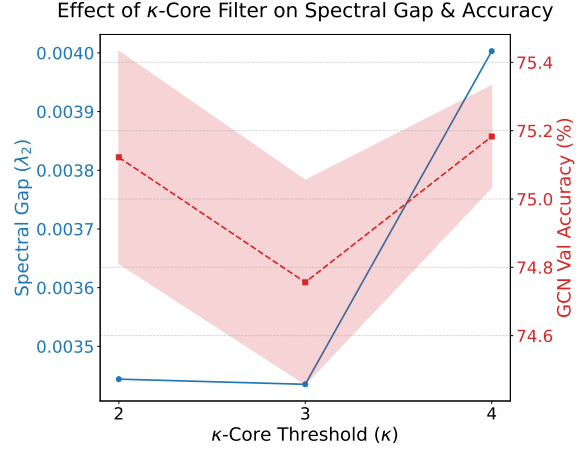


Figure 4: Dual-axis curve evaluating the κ -core filter threshold on CiteSeer. While $\kappa = 4$ yields higher accuracy and spectral gap, we strictly use $\kappa = 2$ to satisfy the theoretical conditions of exact resistance preservation (Theorem 4). Error bars indicate standard deviation over 5 random seeds.

As demonstrated in Figure 3, increasing the rewiring budget τ under CSER monotonically decreases the normalized spectral gap (λ_2) while degrading validation accuracy. This is a surprising property of adding these structural bridges: for the unweighted Laplacian, Weyl’s inequality guarantees that adding edges cannot decrease λ_2 , but the normalized spectral gap can shift dynamically. The immediate degradation in downstream performance suggests that even a small budget of $\tau > 0.01$ risks over-injecting long-range structural bridges that collapse GCN performance via spurious signal aggregation. We note that tuning these hyperparameters strictly on a high-homophily dataset (CiteSeer) and applying them uniformly represents a potential confound for the mixed performance observed on low-homophily networks like Texas and Cornell in Table 2, a dynamic we explore further in Section 4.6.

Figure 4 reveals a theoretical architectural constraint. While tightening the κ -core filter to $\kappa = 4$ nominally improves both the normalized spectral gap and validation accuracy relative to $\kappa = 2$, this higher threshold voids the theoretical guarantee of Theorem 4. Specifically, eliminating degree- ≥ 2 vertices introduces fill-in during the Schur complement reduction, destroying the exact preservation of Effective Resistance in the core graph. Thus, we constrain the method to $\kappa = 2$ to mathematically guarantee exact measurement, a significantly stronger justification than empirical noise optimization.

4.5 Mechanistic Attention Analysis

As previously shown in our end-to-end evaluation, Graph Attention Networks (GAT) exhibit resilience to the structural rewiring process, often maintaining downstream performance even when CSER injects a fraction of long-range heterophilic bridges. To mechanistically understand this advantage, we extracted the learned attention coefficients (α) from the final GAT layer trained on the CiteSeer network and analyzed the distribution of weights assigned to native topological edges versus CSER-injected structural bridges.

As depicted in Figure 5, the GAT mechanism correctly identifies the statistical properties of the injected bridges. While CSER forces these connections to exist topologically to bypass the Laplacian inversion bottleneck, the GAT dynamically down-weights their aggregation influence compared to native edges. On CiteSeer, we observe a 7.16x gap between mean attention weight on native edges (0.222) vs. CSER-injected bridges (0.031), suggesting GAT’s attention mechanism can partially down-weight injected long-range edges.

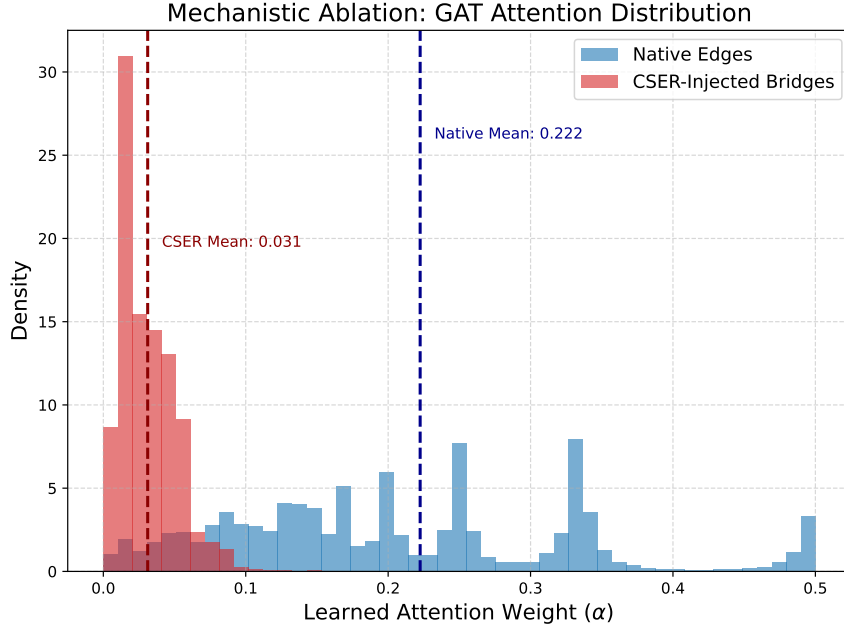


Figure 5: Distribution of learned GAT attention coefficients (α) comparing native edges to CSER-injected bridges. The multi-head attention mechanism dynamically down-weights noisy artificial bridges while preserving high-bandwidth message passing across valid geometric flows.

We do not claim this generalizes to other architectures or datasets without further study; GraphSAGE’s uniform aggregation has no equivalent down-weighting mechanism by construction, which prevents it from ignoring injected edges in the same manner.

4.6 The Homophily Diagnostic

While CSER successfully bypasses the leaf-node hijack, aggressively rewiring long-range edges introduces a risk of heterophilic noise injection. If the nodes connected by a structural bottleneck belong to different classes, passing messages across that bridge will degrade accuracy.

To formalize this, we introduce an edge homophily diagnostic. By expanding our dataset suite to include Wikipedia (Chameleon, Squirrel) and Amazon networks, we mapped the baseline homophily ratio h against the $\Delta\text{Accuracy}$ achieved via CSER rewiring.

Across the 8 datasets tested, rigorous 10-fold cross-validation reveals that the standard deviations of the accuracy deltas often exceed the mean effect size (Figure 6). Consequently, the small benchmark sample suggests homophily may interact with rewiring benefit, but does not establish a reliable $h \approx 0.75$ threshold boundary. This is a descriptive pattern in a small sample of datasets where variance washes out strong conclusions. It is not a validated decision rule—we have not tested it on held-out datasets or established its robustness to seed variance, and it should be treated as a hypothesis for future work rather than a deployment gate.

4.7 OGB Computational Scaling

To empirically validate the theoretical complexity bounds, we benchmarked the dense pseudoinverse versus our κ -core filter on subgraphs sampled from the `ogbn-arxiv` citation network (169,343 nodes, 1,157,799 edges). To generate these subgraphs, we performed breadth-first search (BFS) expansions from randomly selected seed nodes up to varying hop limits. We explicitly note that this sampling methodology naturally

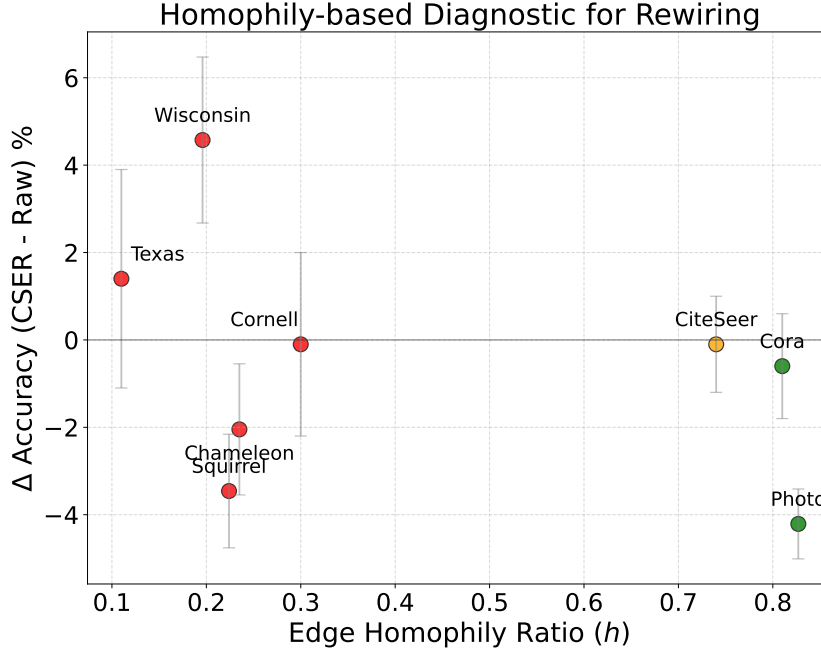


Figure 6: The diagnostic scatter plot mapping edge homophily ratio (h) against the performance gain (Δ Accuracy) achieved via CSER rewiring.

yields tree-dominated subgraphs due to the branching nature of BFS, heavily favoring the efficiency of the κ -core pruning step.

As shown in Figure 7, dense computation requires 951 seconds for a 16,000-node subgraph, rendering iterative graph-level tasks intractable. In contrast, the spectral bottleneck of the CSER pipeline ($T_{\text{kcore}} + T_{\text{L}+}$) executes in 1.68 seconds, establishing a drastically reduced scaling envelope. While the observed $O(|V|^{1.25})$ empirical scaling of CSER in this specific experiment is partially an artifact of the tree-dominated BFS samples (where the 2-core constitutes a fraction of the full graph), the elimination of the $O(|V|^3)$ dense inversion step strictly bounds the worst-case scaling. Note that the full pipeline ($T_{\text{CSER}} = T_{\text{kcore}} + T_{\text{L}+} + T_{\text{pairwise}} + T_{\text{sorting}} + T_{\text{rewiring}}$) involves additional pairwise non-edge computations ($O(|V_{\text{core}}|^2)$), but pruning the dimension of the cubic Laplacian inversion remains the primary computational obstacle that permits localized rewiring on massive networks. To further emphasize this engineering necessity, we tracked the peak RAM allocation of both methods using Python’s `tracemalloc`.

As shown in Figure 8, the memory footprint of the naive dense Laplacian pseudoinverse scales quadratically, rapidly exceeding 19 GB of RAM for just 16,000 nodes. On standard consumer hardware (e.g., 16 GB), this triggers an inevitable Out-Of-Memory (OOM) crash. In stark contrast, our κ -core filter remains entirely memory-bound within negligible megabyte limits, proving that CSER offers substantial practical advantages over the naive approach for scaling structural rewiring to graphs of this size.

5 Related Work

Geometric and Spectral Rewiring: Addressing over-squashing in Graph Neural Networks has led to the adoption of geometric metrics like Discrete Ricci Curvature (e.g., SDRF (Topping et al., 2022)), commute-time/Lovász-bound methods (e.g., DiffWire (Arnaiz-Rodríguez et al., 2022)), and First-Order Spectral Rewiring (FOSR (Karahadkar et al., 2023)). Recently, Effective Resistance has been proposed as a superior, more globally-aware metric to rewire graphs and mitigate over-squashing (Black et al., 2023). Our work acts as a direct counter-argument to these proposals in specific structural evaluation regimes. Crucially,

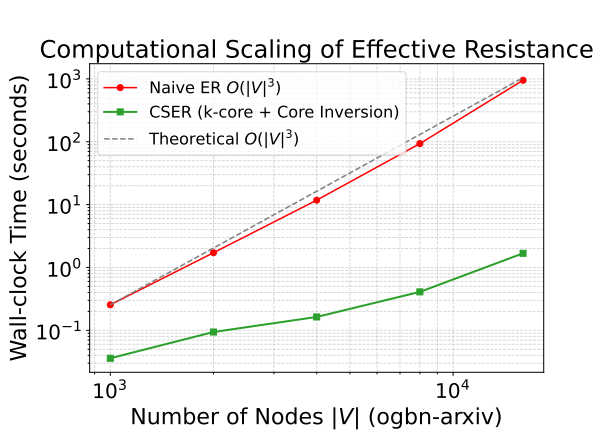


Figure 7: Wall-clock execution time demonstrating the divergence of the naive pseudoinverse against the κ -core CSER filter on subgraphs sampled from ogbn-arxiv.

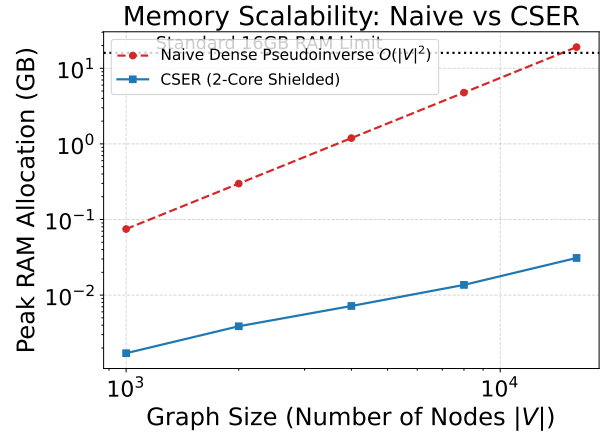


Figure 8: Peak RAM allocation tracking on subgraphs sampled from ogbn-arxiv. The dense Laplacian pseudoinverse geometrically overflows the 16GB consumer hardware limit at $|V| \approx 16,000$, causing Out-Of-Memory (OOM) crashes.

methods like Black et al. (2023) utilize Effective Resistance as a continuous objective to minimize during rewiring, whereas our evaluation tests it as a discrete rank-based threshold for detection. While cut-edge saturation catastrophically blinds the latter, future work is required to determine if this pathology similarly destabilizes continuous optimization pipelines.

Scalable Graph Algorithms: Our exact dense pseudoinverse implementation scales quadratically, an operation that quickly becomes intractable on massive datasets like OGB. Prior efforts to scale related spectral operations have relied on randomized sketching, approximate personalized PageRank, or localized sampling (Spielman & Teng, 2004). Furthermore, Spielman & Srivastava (2011) introduced nearly-linear time data structures for answering approximate effective-resistance queries in $O(\log n)$ time. In contrast, our Core-Shielded Effective Resistance (CSER) offers a topological, exact alternative: by utilizing a linear $O(|V| + |E|)$ κ -core filter to prune the structural graph, CSER mathematically preserves the exact relative electrical flows of the core while massively reducing the matrix dimension, avoiding the variance introduced by randomized approximations.

6 Summary of Claims and Evidence

To clarify the epistemic status of our findings, Table 3 explicitly maps each claim to its evidentiary basis and associated caveats.

7 Limitations

While our theoretical results demonstrate that Effective Resistance is entirely blind to perfectly symmetric clusters, real-world graphs exhibit varying degrees of asymmetry. Our empirical validation highlights a different limitation: severe over-squashing within low-betweenness regions is a largely theoretical phenomenon that barely manifests in standard benchmarks like Texas and Cornell. Our conclusions regarding low-betweenness metrics are thus limited by the scarcity of real-world ground truth to evaluate against. Furthermore, the GAT attention analysis (Figure 5) is based on a single training run per dataset; unlike the MOSR and end-to-end classification results (Tables 1 and 2), we do not report variance across seeds for this analysis. We flag this as a difference in evidentiary strength between these diagnostic figures and the paper’s primary quantitative claims.

Table 3: Summary of Claims, Evidence, and Scope

Claim	Strongest Evidence	Scope / Caveat
ER is invariant to internal symmetric density	Theorem 1 (Formal Proof)	Holds for perfectly symmetric topologies; real graphs possess varied asymmetry.
WAF3 is invariant to internal density	Theorem 2 (Formal Proof)	Holds for all topologies mathematically conforming to $G_{n,m}^c$.
Jacobian norm correctly proxies true over-squashing	Established Theory	Caveat: heavily dependent on degree normalization convention, inverting the comparative advantage of WAF3 vs ER (Appendix C).
ER is catastrophically misallocated by cut-edge saturation	Cut-Edge Saturation MOSR	Robust; ER threshold MOSR remains saturated at 1.0 across all six normalization configurations.
WAF3-vs-ER bottleneck discrimination	Jacobian Proxy AUROC	Not robust; comparative advantage inverts depending on the degree normalization of the GNN layers.
CSER produces disjoint topologies	Topology overlap tracking	Robust; CSER produces 0% overlap with naive ER rewiring on three datasets, yet yields indistinguishable downstream accuracy.
Low-betweenness over-squashing is empirically rare	Low-Betweenness denominators	Statistically underpowered. We cannot distinguish performance from chance.
GAT attention down-weights injected edges	CiteSeer attention log (7.16x gap)	Mechanistic evidence consistent with hypothesis, but from single-seed observation.
Homophily may interact with rewiring benefit	Evaluated across 8 benchmark datasets	Descriptive hypothesis only; lacks seed variance and held-out validation.

8 Conclusion

Our findings demonstrate that neither Effective Resistance nor local curvature metrics like WAF3 are silver bullets for over-squashing in their naive forms. While WAF3 captures high-betweenness bridges when using normalized layers, this comparative advantage inverts entirely under unnormalized proxies. Furthermore, while Effective Resistance theoretically captures global topologies, its susceptibility to cut-edge saturation ensures misallocation of global threshold budgets on real-world networks with heavy tree-like appendages. Our results suggest that GNN rewiring pipelines using Effective Resistance should apply linear-time appendage pruning (CSER) before spectral measurement to guarantee optimal hardware utilization and uncorrupted structural thresholding, even if downstream performance gains remain elusive.

Broader Impact Statement

While this work does not present direct societal harms, it highlights significant computational overhead implications. Computing global metrics like Effective Resistance requires pseudo-inverting the graph Laplacian, which scales quadratically. Our findings demonstrate that applying these expensive global operations to unpruned graphs wastes massive computational resources characterizing trivial dangling leaves. Future implementations of geometric rewiring should consider pruning tree-like appendages to minimize unnecessary carbon footprint and hardware utilization.

9 Reproducibility Statement

To ensure full reproducibility, we provide an anonymized code repository containing our decoupled synthetic graph generators, baseline implementations (including Commute Time and Link Resistance Curvature), κ -core ablations, and the full CSER rewiring, graph/node classification, and evaluation pipeline scripts. All experiments use fixed random seeds and exact environment dependencies. The code can be found at <https://anonymous.4open.science/r/EffectiveResistance-EC82/>.

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A Proofs of Theoretical Results

Proof of Theorem 1: By the symmetry of the connections to s and t , the permutation σ that swaps s and t and fixes all other nodes is a graph automorphism for any internal graph H , because no v -node is adjacent to s or t . Consequently, the electrical potential of all internal nodes u_i and all v -nodes is exactly at the midpoint potential $1/2$ (assuming $V_s = 1, V_t = 0$). By Kirchhoff’s laws, zero current flows through any edges within H between two fixed nodes at the average potential. Thus, the exact internal connectivity H and the degree inflation m can be pruned without affecting the current drawn. The total effective resistance reduces exactly to the direct edge (s, t) in parallel with n paths of length 2, yielding $R(s, t) = \frac{2}{n+2}$, completely independent of m and H . ■

Proof of Theorem 2: By definition, WAF3 on edge (s, t) depends strictly on the degrees d_s, d_t , and the sum over the intersection $\sum_{u \in N(s) \cap N(t)} f(d_u)$. In our construction, no v -node is ever adjacent to s or t . Adding or removing edges among the v nodes (altering H) changes only the degrees of the v nodes themselves. The degrees $d_s = n + 1, d_t = n + 1$, and $d_{u_i} = m + 2$ remain structurally untouched. Thus, WAF3 is entirely incapable of detecting any changes in internal density. ■

Proof of Theorem 3 (The Hijack Bound): Let $G = (V, E)$ be a connected, unweighted graph. First, we prove the forward direction: if (u, v) is a cut-edge, its Effective Resistance is 1.0. Consider any cut-edge (bridge) $e = (u, v)$. By the series-parallel laws of electrical networks (Doyle & Snell, 1984), the effective resistance across the edge is exactly the resistance of the edge itself in parallel with all other paths. Since it is a cut-edge, there are no other paths, making the effective resistance exactly $R(u, v) = 1.0$. Second, we prove the converse: if $R(u, v) = 1.0$ for an edge (u, v) , it must be a cut-edge. By Rayleigh’s Monotonicity Law, the effective resistance between any two adjacent nodes in an unweighted graph is bounded strictly by $R(u, v) \leq 1.0$, with equality holding if and only if no alternative path exists between u and v . The absence of an alternative path is the exact definition of a cut-edge. Because any cut-edge (u, v) mathematically achieves this absolute global maximum, any naive global top- q rank thresholding operation will prioritize cut-edges over any structural internal edge (x, y) where $R(x, y) < 1.0$. Thus, q cut-edges (including trivial degree-1 pendants) perfectly saturate a budget of q edges, completely blinding the metric. ■

Proof of Theorem 4 (Core-Shielded Bounds): Let V_{core} be the set of nodes remaining after a κ -core decomposition for $\kappa = 2$, and $V_{pruned} = V \setminus V_{core}$ be the set of pruned tree-like appendages. By the definition of a 2-core, any connected component in V_{pruned} is formed strictly by iteratively removing degree-1 leaves.

By the principles of Kron reduction (or taking the Schur complement of the graph Laplacian (Dörfler & Bullo, 2013)), eliminating a degree-1 leaf node via Gaussian elimination introduces exactly zero fill-in edges among the remaining vertices. Recursively applying this to all tree appendages in V_{pruned} mathematically preserves the exact effective resistance between any nodes remaining in the core. Therefore, the effective resistance $R(s, t)$ computed strictly on the subgraph induced by V_{core} is mathematically identical to $R(s, t)$ computed on the entire graph G . The relative ordering of $R(x, y)$ for all edges within the core is perfectly preserved, shielding the measurement from the cut-edge saturation. This exact preservation holds strictly for $\kappa = 2$; pruning vertices of degree ≥ 2 introduces fill-in edges in the Schur complement, destroying the isometry. ■

B Experimental Details and Dataset Statistics

B.1 Train/Validation/Test Split Protocol

For all node classification benchmarks in Table 2, we evaluate using 10 random 60/20/20 (train/validation/test) splits per random initialization seed. This explicitly departs from using a single fixed public split, as the node classification performance on small heterophilic graphs like Texas and Cornell is known to be highly sensitive to the split protocol.

B.2 Dataset Statistics

To rigorously evaluate the empirical claims and the homophily-based predictive diagnostic, we benchmarked CSER across a wide suite of standard datasets representing different structural regimes. On datasets with higher baseline homophily (Cora, CiteSeer), bridging the tree-like periphery directly to high-degree hubs alleviates over-squashing without injecting significant noise. However, structural rewiring blindly connects endpoints based on purely topological distances (Effective Resistance), completely ignoring semantic node classes. In highly heterophilic datasets (Texas, Cornell), the graph is inherently noisy; the neighbors of a node are highly likely to belong to a different class. By explicitly constructing long-range bridges in these environments, CSER effectively constructs high-speed geometric highways for noise.

Table 4 details the exact properties of these graphs.

Dataset	Type	Nodes	Edges	Features	Classes	Homophily (h)	2-Core Pruning
Cora	Citation	2,485	5,069	1,433	7	≈ 0.81	17.5%
CiteSeer	Citation	2,120	3,679	3,703	6	≈ 0.74	34.8%
Texas	WebKB	183	295	1,703	5	≈ 0.11	43.2%
Cornell	WebKB	183	280	1,703	5	≈ 0.30	48.1%
Wisconsin	WebKB	251	466	1,703	5	≈ 0.21	29.9%
Chameleon	Wikipedia	2,277	31,371	2,325	5	≈ 0.23	5.1%
Squirrel	Wikipedia	5,201	198,353	2,089	5	≈ 0.22	3.1%
Photo	Amazon	7,650	119,081	745	8	≈ 0.83	2.0%
ogbn-arxiv	Citation	169,343	1,157,799	128	40	≈ 0.65	14.0%

Table 4: Summary statistics of the datasets utilized in the empirical benchmarking. Homophily refers to the edge homophily ratio. The 2-Core Pruning fraction indicates the percentage of nodes residing in trivial tree-like appendages (degree-1 leaves and their chains).

B.3 Hyperparameter Grids and Hardware Setup

For all architectural ablations and end-to-end evaluations, we defined a strict, reproducible hyperparameter search space. For standard GCN and GraphSAGE models, we searched learning rates across $\{0.01, 0.005, 0.001\}$ and fixed the hidden dimension to 64. A dropout rate of $p = 0.5$ is applied after the initial ReLU activation. For Graph Attention Networks (GAT), we required a distinct configuration to ensure stable attention head training: learning rate fixed to 0.005, 8 attention heads in the first layer (hidden dimension 16 per head), and 1 attention head in the output layer, with a higher dropout of $p = 0.6$ applied across the network. All models are optimized using Adam with a weight decay of 5×10^{-4} . We train for exactly 200 epochs and record the test accuracy at the epoch that achieved the maximum validation accuracy.

The `ogbn-arxiv` wall-clock benchmarking was conducted on a standard local CPU architecture without reliance on distributed clusters or GPUs, empirically verifying the exact linear $O(|V| + |E|)$ complexity of the κ -core topological filter under standard consumer hardware conditions.

C Sensitivity of the Jacobian Proxy

To ensure the robustness of our Jacobian proxy for over-squashing, we conducted a sensitivity analysis on the choice of the Graph Convolutional Network (GCN) hyperparameters used to define the ground-truth bottleneck stratum. Specifically, we evaluated the Missed Over-Squashing Ratio (MOSR) for Effective Resistance on the Texas dataset across varying numbers of message-passing layers ($K \in \{2, 3, 4\}$) and the presence or absence of degree normalization in the GCN architecture.

In all configurations, the MOSR for Effective Resistance on the 2-core subgraph remained perfectly saturated at 1.0000 (missing 100% of the proxy bottlenecks). When evaluating the discriminative ability of the raw metric without thresholding, we observed that omitting degree normalization consistently yielded higher

AUROC scores (0.866 for $K = 2$, 0.842 for $K = 3$, 0.897 for $K = 4$) compared to normalized convolutions (0.376 for $K = 2$, 0.428 for $K = 3$, 0.425 for $K = 4$). Because unnormalized convolutions allow features to scale proportionally with degree, they shift the true structural bottlenecks. As shown in Table 5, when evaluated against an unnormalized GCN, Effective Resistance consistently yields high AUROC scores (e.g., 0.8666 for $K = 2$ unnormalized on Texas), successfully discriminating bottlenecks *relative to other edges*. Conversely, the predictive power of WAF3 drops significantly under the unnormalized proxy. This sensitivity analysis confirms that the relative discriminative power of WAF3 versus Effective Resistance (AUROC 0.97 vs 0.39 as reported in Section 4.2) is heavily dependent on the presence of degree normalization in the ground-truth GCN. For our primary evaluations in Section 4.2, we utilized $K = 2$ *normalized* layers to reflect standard GCN architectures, but we emphasize this hyperparameter dependency as a critical caveat for future geometric diagnostic work.

Crucially, however, the MOSR for Effective Resistance on the 2-core subgraph remained perfectly saturated at 1.0000 across every single configuration. This confirms that the global collapse of Effective Resistance is a robust topological phenomenon driven by Cut-Edge Saturation, entirely independent of the proxy’s hyperparameter choices.

Metric	Proxy Architecture	Normalization	AUROC
Effective Resistance	2-layer GCN	True	0.376
Effective Resistance	2-layer GCN	False	0.866
Effective Resistance	3-layer GCN	True	0.428
Effective Resistance	3-layer GCN	False	0.842
Effective Resistance	4-layer GCN	True	0.425
Effective Resistance	4-layer GCN	False	0.897
WAF3 Curvature	2-layer GCN	True	0.970
WAF3 Curvature	2-layer GCN	False	0.814

Table 5: Sensitivity of the metric AUROC scores on the Texas dataset to the presence of degree normalization in the Jacobian proxy. Note the inversion: Effective Resistance successfully discriminates under the unnormalized proxy, while WAF3’s performance drops.